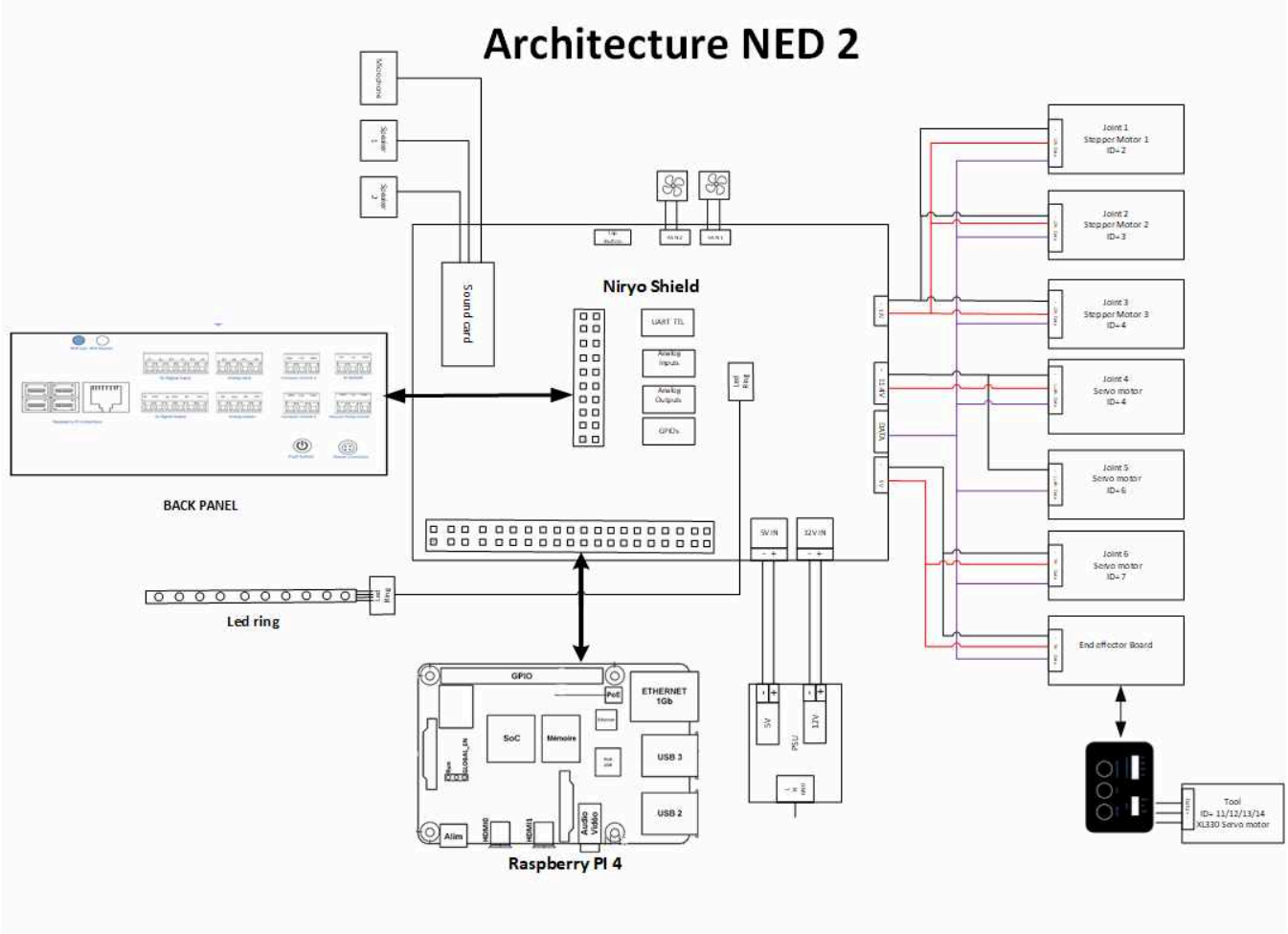
Electrical interface

Electrical interface of Ned2

Electrical interface overview



Back Panel interface

Ned2 comes with I/O interface that can be used for wide range of equipements and Ned2 ecosystem.

The illustration below shows the differents electrical and communication interfaces present of the back panel of the robot.



1	2 x USB 2.0
2	2 x USB 3.0
3	RJ45 port
4	Wi-Fi status indicator
5	Wi-Fi button
6	3 x digital inputs
7	3 x digital outputs
8	2 x analog inputs
9	2 x analog outputs
10	Conveyor control interface
11	Conveyor control interface
12	Infrared sensor interface
13	Vacuum pump control interface
14	On/off button
15	General power input

⚠Take great care in following the safety guidelines in the next section to avoid damaging the robot.

Be sure to respect the order of connection and the characteristics of the robot's inputs/outputs.

Make sure all the equipment and cables are dry. If water gets into the equipment, unplug the power cable.

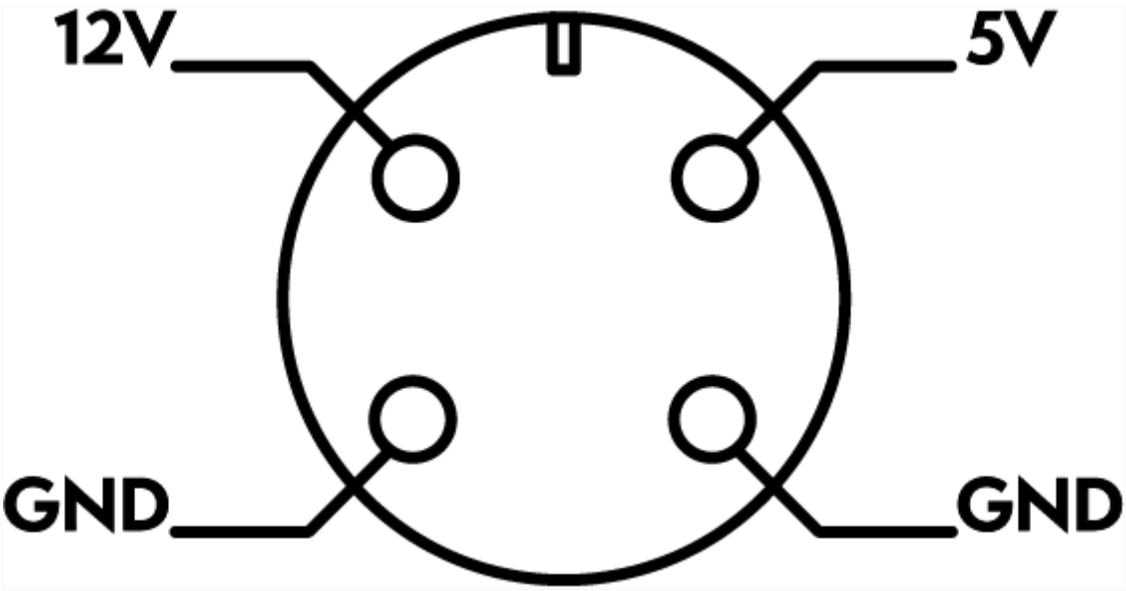
Do not unplug or turn off the robot while it is in use, as this could damage the robot's Raspberry Pi 4 or the micro SD card inside.

Before plugging in or unplugging the power supply, be sure to turn off the robot.

General power input

Ned2 is powered with AC/DC adapter with two output ratings: 12V and 5V

Settings	Value
Frequency	50/60 Hz
Input	AC 100-240 V , 2.5 A
Output	12 V – 7 A; 5 V – 7 A



On/Off button

The robot is **turned on** by pressing the power button on the back panel.

When the robot is starting on , the led ring displays white light.

After the robot start up completed, the indication of led ring will display blue light, the robot can be used normally at this time.

> To avoid any connection problem, be sure to wait for the blue Led before connection or using the robot.

> Make sure the robot is in home position when powering off

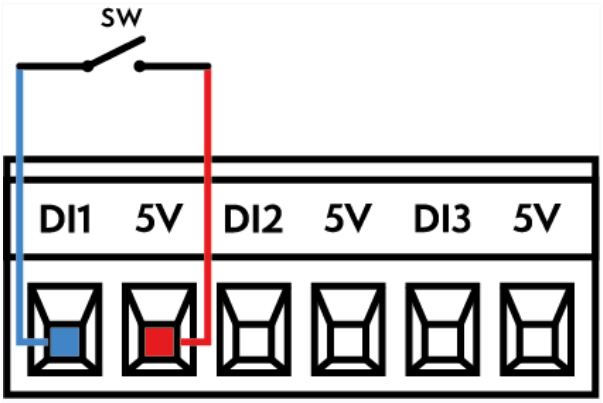
Digital inputs

The digital input are NPN types.

The electrical specifications for the connectors [DI1 - DI2 -DI3] are shown below:

Settings	Min	Max	Units
Voltage	0	5	V
Current	0	5	mA

This illustration below shows how to connect a switch to a digital input.



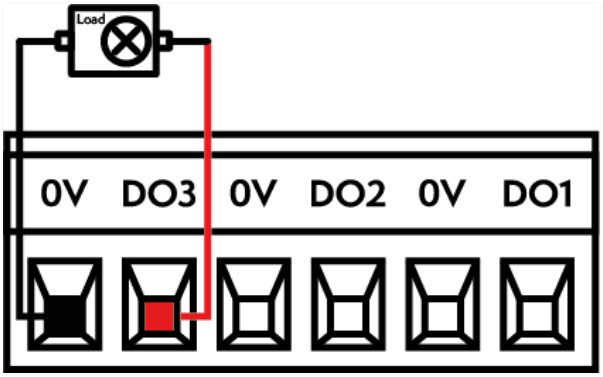
Digital outputs

The digital output are PNP types

The electrical specifications for the connectors [DO1 - DO2 -DO3] are shown below:

Settings	Min	Max	Units
Voltage	0	5	V
Current	0	250	mA

This example shows how to connect a load in order to be controlled from a digital output.



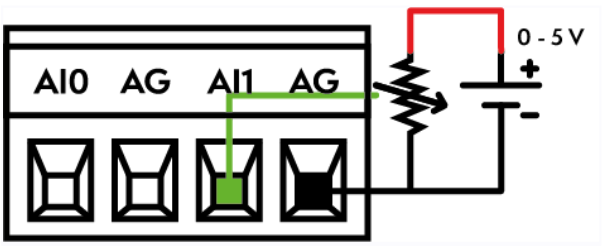
Analog inputs

The analog inputs can be used to measure voltage (0-5V) form an external equipement.

The electrical specifications for the connectors [AI1-AI2] are shown below:

Settings	Min	Max	Units
Voltage	0	5	V
Current	0	20	mA
Resolution		12	bit

This illustration below shows how to connect a potentiometer for reading values from 0-5V.



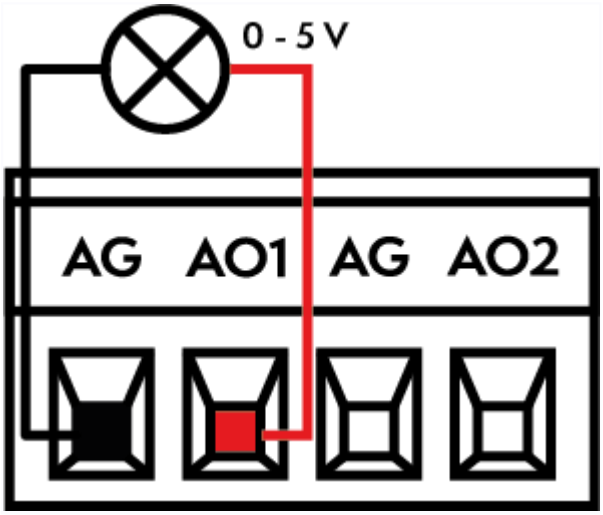
Analog outputs

The analog outputs can be used to set voltage (0-5V) to an external equipement.

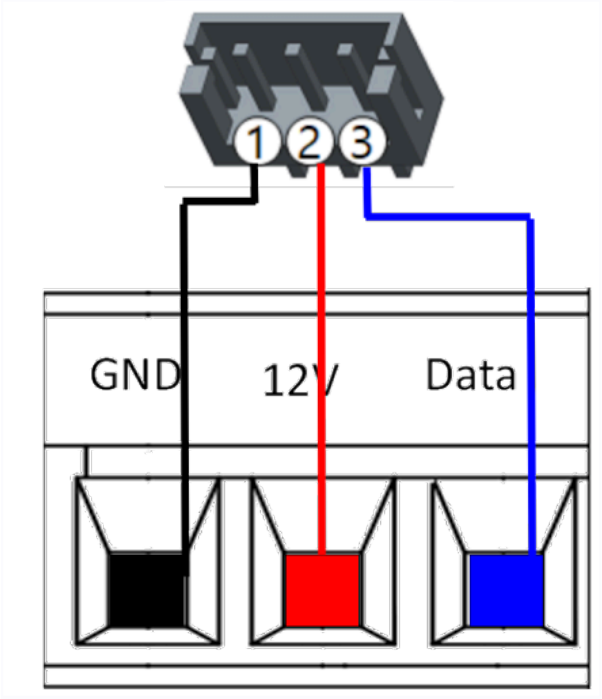
The electrical specifications are shown below:

Settings	Min	Max	Units
Voltage	0	5	V
Current	0	30	mA
Resolution		12	bit

This example shows how to connect a load in order to be controlled from an analog output.

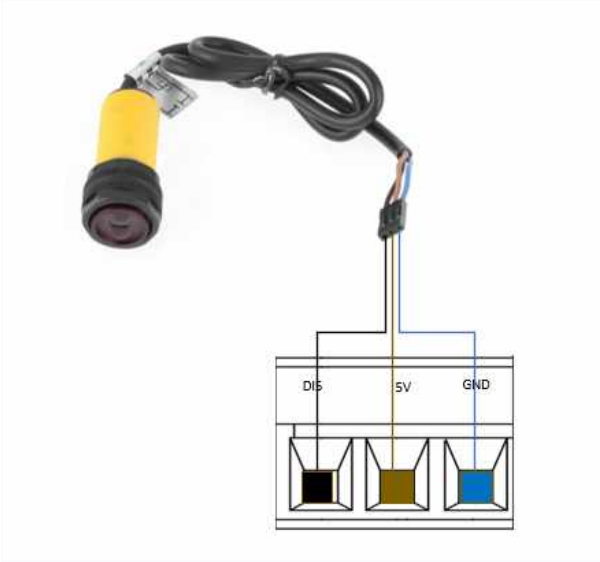


Conveyor control interfaces

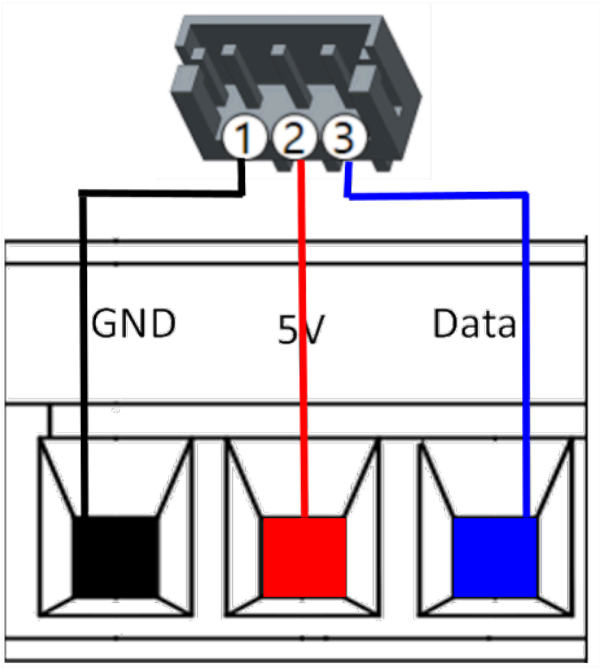


Please refer to the conveyor v2 documentation for more information

Infrared sensor interface



Vacuum pump control interface



Hotspot mode

The Wi-Fi indicator is used to display the Wi-Fi status in the robot.

Color	Description	Trigger
On	Hotspot is on and available	Hotspot activated using Niryo Studio or the Wi-Fi button
Off	Hotspot is off and unavailable	The robot is connected to the network either with an Ethernet cable or via Wi-Fi

Hotspot button

Function	Type of action	Condition	Status after action
Activate the hotspot	Short press	The robot's hotspot is Off	The robot's hotspot is On
Deactivate the hotspot	Short press	The robot's hotspot is On	The robot's hotspot is Off
Reset the hotspot	Long press	Any	Thhe hotspot is reset

Top button

The top button on the Ned2 has multiple functions, depending on how long your press on it or the number of presses.

Time/Number of presses	Function
1 short press (0.1 and 2 seconds)	Play the saved demo program | Pauses/Resume the program running
1 long press (> 2 seconds)	Stops the program

Control Panel



FreeMotion button

Pressing and holding this button deactivates the torques in the motors and performs the FreeMotion operation (find out more information on operation mode section.

Save button

A short press on this button saves the robot's position on NiryoStudio

A long press will save a the trajectory made while holding on NiryoStudio

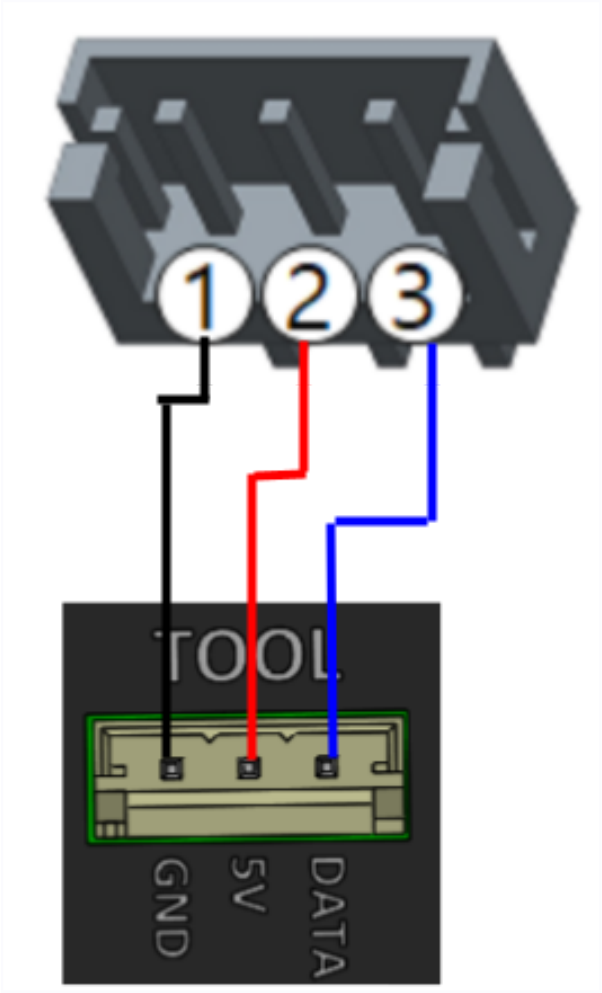
Custom button

This button is programmable and allows the user to choose the type of action and the function of the button.

Check the dedicated API sections to use it in your programs.

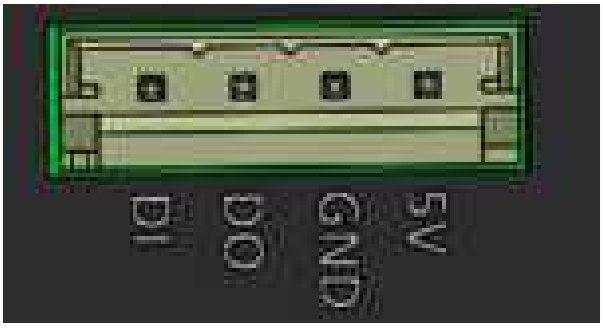
Gripper connector

This connector is used to connect and control the Ned2's gripper.



Electromagnet connector

This connector is specifically for connecting Niryo's electromagnet.



This connector can be use to connect other element than the electromagnet is needed.

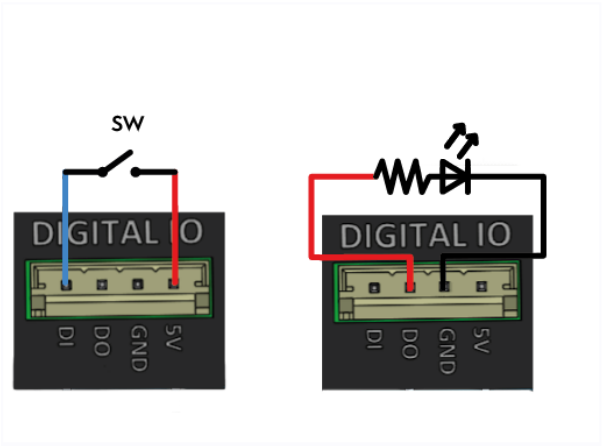
Digital output Type->PNP:

Connector	Settings	Min	Max	Units
[DO4]	Voltage	0	5	V

Connector	Settings	Min	Max	Units
[DO4]	Current	0	250	mA

Digital input Type->NPN:

Connector	Settings	Min	Max	Units
[DO4]	Voltage	0	5	V
[DO4]	Current	0	5	mA



Safety box

The safety box contain the emergency stop. Activate the emergency stop button to immediately stop all robot motion.

⚠ The safety bow should always be wired between the alimentation and the robot.



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